MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOMULTIPLIER

TYPE 5819

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: 10-stage, head-on, S11 spectral response

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ebb	Eb-Edy10	Light source	Light	Ib	TA	Rp	Cathode area Lighted	Alt
Unit:	Vdc	Vdc	---	lm	mAdc	°C	Ohms	sq in.	ft
Maximum:	1,250 (Note 18)	250 (Notes 1 and 18)	---	---	0.75 (Note 2)	75	Note 3	---	40,000
TEST CONDITIONS:	1,000 (Note 4)	Note 4	LS1	10^{-5}	---	---	10,000	0.12	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) - 336 hours

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
Ⓒ ---	Low-pressure breakdown	4, 17, 19, 20	Ebb = 1,250 Vdc; Eb to Edy 10 = note 18; Rp = 1 Meg; t = 120	---	---	---	---	---	---
1331	Direct-interelectrode capacitance (1)	13		---	---	C	2.7	5.7	pF
1331	Direct-interelectrode capacitance (2)	14		---	---	C	5.0	8.0	pF
Ⓒ 5903	Spectral response identification	-	S11	---	---	---	---	---	---
Ⓒ 1216	Base material insulating quality	-		---	---	---	---	---	---
	Quality conformance inspection, part 1	21							
---	Cathode current	6, 7	Light = 0.01 fc	---	---	Ik	0.40	---	μAdc
5301	Dark current (1)	8		---	---	Ldb	---	0.04	μAdc
5301	Anode current	6	Rp = 10 K	---	---	Ib	100	---	μAdc
---	Anode current ratio	9	Rp = 10 K	---	---	IbR	---	3.0	---
5301	Dark current (2)	10	Light = 0	---	---	Ldb	---	5.0	μAdc
---	Blue sensitivity	6, 7, 11		---	---	Ik	0.040	---	μAdc
5306	Signal-to-noise ratio	12	Light = 10^{-7} fc Rp = 0.1 Meg	---	---	S/N	42	---	dB
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	5		---	---	---	---	---	---
Ⓒ 1111	Base pin solder depth	-		---	---	---	---	---	---
Ⓒ 1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	15	Group D; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
	Anode current	16		---	---	---	---	---	---

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NOTES:

1. DC supply voltage between anode and 10th dynode.
2. Averaged over a period of 30 seconds maximum. Longer tube life may be expected when operating conditions require lower anode current.
3. An anode load resistance of at least 10,000 ohms is recommended for a protective resistance.
4. Overall supply voltage is across a voltage divider providing 1/6 of the voltage between cathode and dynode No. 1 and 1/12 of the voltage for each succeeding stage including dynode No. 10 to anode.
5. Omit immersion in water and test dry. Apply torque test with 40.0 pound-inches minimum.
6. Light source shall be a projection lamp operated at 2,870°K. Use a light aperture 0.172 inch x 0.70 inch (4.37 mm x 17.78 mm) centered with respect to the cathode.
7. Apply 167 volts dc between cathode and all other pins tied together.
8. With 10^{-5} lumens incident on the cathode vary the overall voltage to give Ib = 200 μA; exclude the light and read the dark current.
9. Anode current ratio. In addition to Ib test at center of photocathode, check Ib at four places spaced 90° apart. The spots shall be exposed one at a time in such a manner that no overlap of outer spots exist (if all were exposed at the same time). The ratio between maximum and minimum anode currents (including center reading) will be the anode current ratio. Place aperture between light source and tube such that the outer spots (on the photocathode) will lie on a 7/8 inch ± 1/32-inch (22.23 mm ± 0.79 mm) diameter circle centered on the photocathode. The diameter of the outer spots on the photocathode shall be 1/2 inch (12.7 mm) ± 5 percent. All spots including center spot shall have equal area.
10. Apply Ebb according to the range of Ib in the following table:

	Ib	Overall supply voltage
Range	100 to 160	1,350
	160 to 240	1,300
	240 and up	1,250

11. With unfiltered light flux of 0.01 lumens, insert blue filter (Corning, Glass Code No. 5113 polished to 1/2 stock thickness, or equivalent) in place between the cathode and light source.
12. For this test, the light source shall be a square wave modulated at 90 Hz. Diameter of the light spot shall be 1-1/4 inches (31.75 mm). Output of the tube shall be fed into a low-pass amplifier, the bandwidth of which is 1,000 Hz. The amplifier output shall be read with a vu meter calibrated in dB. Amplifier attenuator is also calibrated in dB. Prior to testing, tubes shall be held at an ambient temperature of 75° ± 5°F for not less than 1 hour in the dark.

    (a) Signal - with light on, adjust attenuator so that vu meter is at zero. Read attenuator setting.

    (b) Noise - read as above with light off.

    (c) S/N Ratio = Signal (dB) - Noise (dB).

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is provided for obtaining information on the use of this specif

NOTES: - Continued

13. Capacitance between anode and dynode No. 10, all other elements grounded.

14. Capacitance between anode and all other elements.

15. Ebb = 1,000 volts, adjust luminous flux at the cathode initially to obtain Ib = 0.75 mAdc. Color temperature shall be greater than 2,000°K.

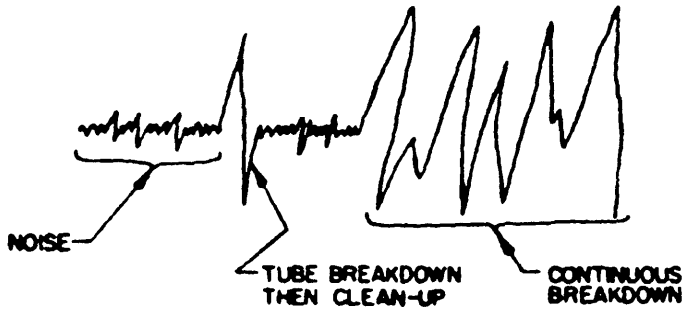
16. Life-test end-point is either an Ib of 20 μa or a decrease to 1/20 of the initial Ib, whichever is reached first.

17. Test circuit and test equipment requirements.

A bleeder network to provide 1/6th of the overall voltage between cathode and dynode No. 1 and 1/12th of the overall voltage for each succeeding stage; 8-megohm series resistance for cathode, and all dynodes excluding the last dynode (10th); 1-megohm series resistance for last dynode (10th) and anode; a regulated (0.1 percent) power supply with low-ripple (0.05 volts peak-to-peak maximum) as a voltage source; a sensitive dc microammeter to determine anode current; an oscilloscope, with a low-capacity divider, is connected across the anode load resistance (1-megohm) to monitor noise and breakdown; base pins of photomultiplier under test connected so that the degradation of breakdown characteristic is minimized; a vacuum pump, low-pressure chamber and manometer; a light source with an adjustable iris and sandblasted glass.

Test method.

Mask tube with light proof material. The light-source should be located so that light from it is striking the cathode of the photomultiplier. (Light from the light source should be the only light striking the cathode.) The opening of the iris is varied to obtain approximately 10 microamperes of anode current. Observe the tube-noise signal as displayed by the oscilloscope. Reduce pressure of vacuum chamber to 130 millimeters of mercury. The tube should be subjected to this pressure for a period of 2 minutes. During pressure reduction, the tube noise signal, as displayed by the oscilloscope, should be observed to determine tube breakdown. When breakdown occurs, the signal being monitored by the oscilloscope will resemble the following:



When continuous breakdown occurs, a change in anode current will result.

18. An ac supply voltage may be used having a peak value equal to this maximum value.

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NOTES: - Continued

19. Vary light to obtain approximately 10 microamperes of anode current.

20. Tube shall operate without indication of breakdown in an evacuated chamber in which the pressure does not exceed 130 mmHg absolute.

21. All quality conformance inspection, part 1, tests to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11
DSA - ES

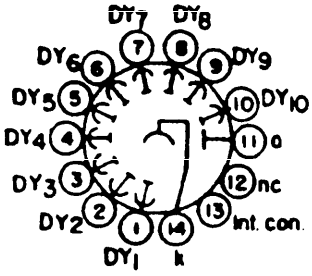
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

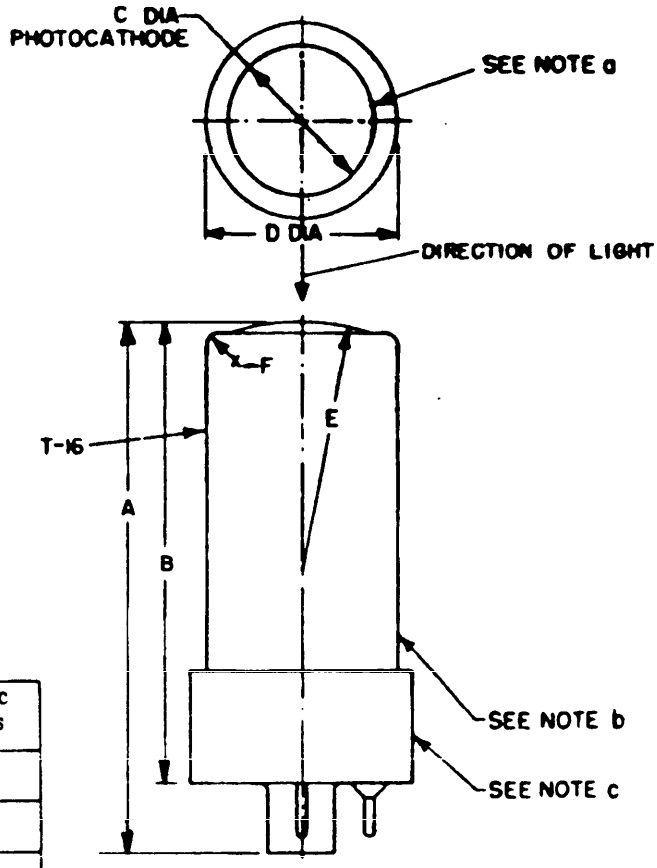
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2556)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
C	1.688 (42.88)	
E	3.000(76.20) R	
F	0.150 (3.81) R	
Quality conformance inspection, part 2		
A	5.375(136.53)	5.875(149.23)
B	4.625(117.48)	5.125(130.18)
D	1.938 (49.23)	2.063 (52.40)

NOTES:
a. Diameter C refers to the cathode diameter.
b. Centerline of bulb shall not deviate more than 2° in any direction from the perpendicular erected at center of bottom of base.
c. The base shall be a medium shell diheptal 14-pin base, B14-38.

FIGURE 1. Outline drawing of electron tube type 5819.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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